

IIT Placement Intelligence Platform

Software Requirements Specification (SRS)

1. Project Overview

Project Name

IIT Placement Intelligence Platform

Problem Statement

Placement information across IITs is distributed across annual reports, placement brochures, RTI responses, and news articles. Students often struggle to compare institutes, branches, and placement outcomes using reliable data.

The IIT Placement Intelligence Platform centralizes placement statistics from multiple IITs and provides interactive analytics, visualizations, comparisons, and predictive insights to help students make informed academic and career decisions.

2. Target Users

Primary Users

- JEE aspirants comparing IITs and branches
- Students exploring placement trends
- Academic researchers interested in placement analytics

Secondary Users

- Placement cells
 - Career guidance communities
 - Educational analysts
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3. Objectives

The platform should:

- Aggregate placement statistics from multiple IITs.
- Enable branch-wise and institute-wise comparisons.
- Provide trend analysis across years.
- Analyze sector-wise recruitment patterns.
- Predict future placement outcomes using machine learning.

- Present insights through an intuitive dashboard.
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4. Technology Stack

Frontend

- React
- Tailwind CSS
- Recharts

Backend

- FastAPI (Python)

Database

- MySQL

Machine Learning

- Pandas
- NumPy
- Scikit-Learn
- Random Forest Regressor

Deployment

- Frontend: Vercel
- Backend: Render
- Database: MySQL Cloud Instance

Version Control

- Git
 - GitHub
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5. Functional Requirements

FR-1: Dashboard

Display:

- Total IITs covered
- Years covered
- Average package summary
- Placement percentage summary

- Top analytical insights
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FR-2: IIT Comparison

Users should be able to:

- Select multiple IITs
 - Compare placement statistics
 - Compare average packages
 - Compare placement percentages
 - Compare highest domestic packages
 - Compare highest international packages
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FR-3: Branch Analytics

Users should be able to:

- Analyze branch-wise placement trends
 - Compare branches across institutes
 - Identify highest-paying branches
 - Identify highest placement-rate branches
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FR-4: Sector Analytics

Users should be able to:

- Analyze recruitment sectors
 - Compare IT, Core, Finance, Consulting, and other sectors
 - View sector-wise placement distribution
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FR-5: Trend Analysis

Users should be able to:

- Analyze historical package growth
 - Analyze historical placement percentage growth
 - View institute-wise trends
 - View branch-wise trends
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FR-6: Placement Prediction

Users should be able to input:

- Institute
- Branch
- Program
- Year

The system should predict:

- Average package
 - Placement percentage
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6. Machine Learning Module

Model 1: Average Package Prediction

Input Features:

- Institute
- Branch
- Program
- Year

Output:

- Predicted Average Package

Algorithm:

- Random Forest Regressor
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Model 2: Placement Percentage Prediction

Input Features:

- Institute
- Branch
- Program
- Year

Output:

- Predicted Placement Percentage

Algorithm:

- Random Forest Regressor
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7. Database Design

Table: institutes

Fields:

- id
 - institute_name
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Table: placement_stats

Fields:

- id
 - institute_id
 - year
 - program
 - branch
 - avg_package
 - median_package
 - highest_domestic_package
 - highest_international_package
 - placement_percentage
-

Table: sector_stats

Fields:

- id
 - institute_id
 - year
 - branch
 - sector
 - students_placed
 - placement_percentage
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8. Key Insights Generated

Examples:

- Highest-paying IIT in a selected year
 - Highest-paying branch across all IITs
 - Fastest-growing branch
 - Highest placement percentage
 - Most dominant recruitment sector
 - Year-over-year salary growth
 - Institute ranking based on selected metrics
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9. Non-Functional Requirements

Performance

- Dashboard response under 2 seconds

Scalability

- Support addition of new IIT datasets

Reliability

- Handle missing values gracefully

Usability

- Mobile-responsive interface
 - Clean modern dashboard design
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10. Future Scope

- Internship analytics
 - Company-wise recruiter analytics
 - Placement forecasting for future years
 - NIT and IIIT integration
 - Advanced ML models
 - Recommendation system for branch selection
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11. Expected Outcome

A production-ready web platform that helps students compare IITs, analyze placement trends, and obtain data-driven insights for academic and career decision-making.